

SIDS: A federated learning approach for intrusion detection in IoT using Social Internet of Things

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ARTICLE INFO

Keywords:

Federated learning
IoT
SIoT
Privacy
Security
Generative Adversarial Networks

ABSTRACT

The Internet of Things (IoT) ecosystem needs Intrusion Detection Systems (IDS) to mitigate cyberattacks and exploit security vulnerabilities. Over the past years, utilizing machine learning in IDSs has gained a lot of attention. However, in many current works, the training data from different locations should be collected in a central server to be used in the learning process. This data-sharing procedure increases concerns regarding data privacy and decreases the data holders' motivation to participate in the learning process. The use of distributed learning models has been considered a solution to overcome concerns related to privacy. However, these distributed learning models are vulnerable in the presence of untrusted nodes that can participate in the learning process and deteriorate performance. In this paper, we propose SIDS (Social Intrusion Detection System), a trust-oriented federated learning approach for intrusion detection in IoT that utilizes the Social Internet of Things (SIoT). The proposed approach leverages the social relationships among the objects in a system to provide a privacy-preserving collaborative mechanism for detecting intrusions in IoT environments. The experimental results show the proposed solution outperforms the learning models on individual servers while providing a privacy-preserving and trustable environment for collaboration.

1. Introduction

The Internet of Things (IoT) is a system of interconnected devices that can exchange data over a network. Because the devices and data in the IoT ecosystem are expected to be remotely accessible, it opens opportunities for intruders to attack the resources in IoT. To safeguard IoT systems against adversaries, the use of Intrusion Detection Systems (IDSs) has been explored in the past. An IDS monitors the network traffic data to detect suspicious activities, known attacks, and malicious actions [1,2].

To provide dynamic adaptive intrusion monitoring in IoT, machine learning has been widely used in various solutions [3,4]. Although machine learning-based solutions have shown a great ability for dynamic and accurate attack intrusions in IoT, some concerns have been raised regarding data privacy in the learning process. Many of the existing IDS that utilizes machine learning are centralized which means massive, including sensitive and personal data, data from different IoT devices have to be uploaded to central servers. This approach puts the privacy of the transmitted data at risk which raises data security and privacy threats and decreases the motivation of data holders to

participate in the learning process [5]. The other disadvantage of central IDSs is that a great deal of data has to be processed on a server and it is computationally expensive. Federated learning can be a potential solution to address these problems. This approach eliminates the need for data to be collected in a central unit by distributing the learning process among different participants [6]. Using federated learning, data remain private on the local servers and a copy of the learning model is trained over these data, then only the weights of the trained model are transferred to a central server. This approach decreases the workload of the central server and ensures the privacy and confidentiality of the local data. Federated learning is well-aligned with the distributed nature of IoT, especially, when IoT environments leverage edge servers. The edge servers provide enough computation resources for performing and processing tasks at the edge of the network [7].

Despite the advantages of federated learning for privacy-preserving machine learning [8], it brings some challenges in uncertain environments. An uncertain environment is a system in which the participants are not necessarily trusted and there is a possibility that some participants intend to do malicious actions [9]. Different types of attacks can be launched against federated learning architecture including data

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<https://doi.org/10.1016/j.comnet.2023.110005>

Received 3 May 2023; Received in revised form 26 July 2023; Accepted 28 August 2023

Available online 10 September 2023

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poisoning attacks and model poisoning attacks [10]. These attacks are mainly designed to sabotage the generated model. Because the local data are private in the learning process, there is no way for the central server to audit the local models and detect the poisoned models. Although federated learning reduces privacy concerns in the learning process, further studies show that communication costs and long training time are major drawbacks of this approach [11,12]. To address this problem, feature engineering methods have gained attention. These methods can be utilized to reduce the feature space size and accelerate the learning process [13].

With these problems in mind, in this study, we propose SIDS, a trust-based federated learning approach for intrusion detection in IoT that employs the notion of the Social Internet of Things (SIoT). Social IoT, also known as SIoT, is a concept that integrates the Internet of Things (IoT) with human social networks to create an imitation of social networks between humans and objects [14]. The SIoT network allows smart devices to establish social relationships with each other, similar to how humans establish social friendships [15]. This concept considers objects like humans, intelligent and social, as they create their social network to achieve common goals, such as improving functionality, performance, and efficiency, and satisfying their required services [16,17]. The SIoT concept adapts a service-oriented architecture where heterogeneous IoT devices interact and create relationships with each other to achieve a common goal [18]. In the SIoT environment, can select their friend devices based on their previous experience of direct cooperation as well as the reputation of other devices in the environment. Therefore, in SIoT, friend devices are considered more trustable, and it is encouraged to collaborate with and request services from these devices [19,20]. In this study, we introduce a federated learning approach that reduces the risk of malicious actions in the federated learning process using SIoT. Another challenge in federated learning-based solutions is timely convergence due to potential delays in the local training process and network communication constraints. One solution to address this issue is using feature selection to reduce the feature dimensionality and accelerate the learning process. Feature selection in the federated learning process is a challenging task since we can not access all the data to select the most relevant features. Moreover, the data distribution over different local datasets is heterogenous which means the data can differ in terms of distribution, quality, and size. This paper presents a feature selection approach leveraging the SIoT concept which provides an algorithm to find the most useful features based on recommendations from different participants.

In this study, first, we design a GAN-based model poisoning attack to demonstrate the vulnerability of federated learning in presence of untrusted participants. This attack shows the benefits of the presented approach in providing a trust-based collaborative learning process. Using a GAN-based poisoning attack, an adversary can generate fabricated training data to mislead the learning model and reduce the model's performance [21]. In some applications, such as image processing, it has been shown that this attack is highly effective in compromising the final generated model in the federated learning architecture [22,23], but to the best of our knowledge, this is the first attempt to design a GAN-based poisoning attack against intrusion detection systems in IoT. After analyzing the vulnerability of the federated learning approach in uncertain environments, we propose a trust-based federated approach for intrusion detection in IoT. The proposed approach in this study facilitates cooperative privacy-preserving intrusion detection in IoT while preventing untrusted participants from performing malicious actions in the learning process. In the next step, we design a feature selection method that accelerates the learning process. The presented method is a collaborative feature selection approach that leverages cooperation from trusted participants to choose the more useful features for the learning process.

The major contributions of this work can be summarized as follows:

- We design a model poisoning attack using GAN on federated intrusion detection systems. This attack highlights the vulnerability of the federated learning approach, especially in uncertain environments.
- We propose a robust federated approach for intrusion detection in IoT, leveraging Social IoT and incorporating a trust-oriented mechanism for federated learning. This approach provides resilience against compromising attacks, providing a high level of system reliability.
- To further enhance the performance of federated intrusion detection in IoT, we present a collaborative feature selection method. This method effectively reduces system overload and streamlines the learning process, leading to improved overall efficiency.

The rest of the paper is structured as follows. Section 2 provides an overview of the related work. In Section 3, we illustrate the designed GAN-Based poisoning attack in federated learning architecture. Section 4 presents the proposed approach with a detailed illustration of the algorithms. In Section 5 the dataset, metrics, and evaluation of the proposed model are provided. Discussion is presented in Section 6. Finally, Section 7 concludes the paper.

2. Related work

IDSs are crucial to prevent malicious actions in IoT environments. An IDS collects data and logs from diverse activities in a system and analyzes these data to find possible intrusions. Zarpelão et al. [1] reviewed different studies on IDSs in the IoT area and discussed the major challenges in this field. Recently, with the rapid progress of AI-based technologies, machine learning has been widely used for detecting intrusions in IoT. For instance, Verma and Ranga [24] examined different machine learning techniques for detecting Denial of Service (DoS) in IoT. Larriva-Novo et al. [25] investigated various preprocessing methods for training data to be used in a neural network model. In this article, the authors proposed a new traffic characterization technique for network data. Smys et al. [26] suggested a hybrid convolutional network model for attack detection in IoT. The proposed architecture in this study also leverages a Long Short-Term Memory (LSTM) model for feature extraction in the preprocessing phase of the learning process. Iwendi et al. [27] proposed an ensemble classifier for attack detection systems. The presented architecture classifies the normal and malicious activities utilizing an ensemble model.

Recently, some studies have leveraged the Federated Learning (FL) approach for intrusion detection in IoT. Federated Learning [28], introduced by Google in 2016, is designed to provide an efficient distributed privacy-preserving approach for machine learning. An overview of the federated learning architecture is depicted in Fig. 1. In federated learning, a participant node can be a device with sufficient computation and storage resources (e.g., a laptop or a smartphone) or it can be a middleware that connects the IoT device with limited resources to the network (e.g., an edge server). This architecture is well-aligned with edge-enabled IoT technologies that leverage local servers at the edge of the network. These servers perform some computations locally which leads to reducing in network bandwidth and accelerating communication. As it is shown in Fig. 1, in federated learning architecture every participant trains a local machine learning model using local training data. The model's configuration is determined by the main server and is the same among all participants. After each round of training, the participants send the local trained models to the main server where these models are aggregated to build a global model. Then, the main server broadcasts this global model to all participants and every participant continues the learning process utilizing this model. This procedure continues for several rounds to reach a final global model.

While federated learning has made significant strides in addressing privacy concerns, it does face challenges that can hinder its scalability and efficiency. Communication costs and long training times have been

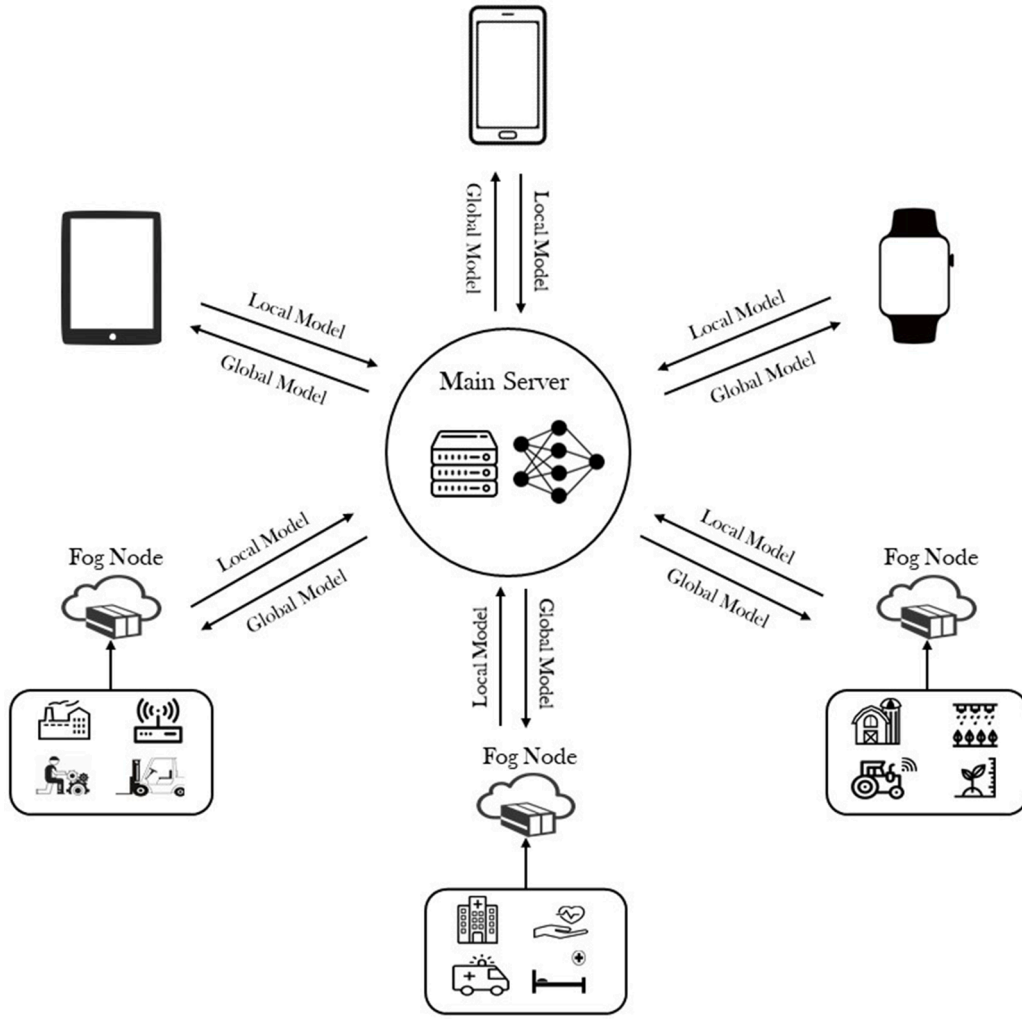


Fig. 1. Federated learning architecture.

identified as major drawbacks of this approach [6]. These limitations can impact the overall performance and effectiveness of the federated system. However, despite these challenges, federated learning remains a valuable and worthwhile approach for several compelling reasons. Firstly, it allows for the utilization of distributed data sources without the need for data centralization, preserving the privacy and security of sensitive information [29]. Secondly, federated learning enables leveraging diverse datasets from different participants, leading to improved model generalization and robustness. This collaborative approach fosters knowledge sharing and facilitates the development of more comprehensive and accurate machine learning models [30]. Lastly, this approach empowers data owners to maintain control over their own data while still contributing to collective intelligence and advancements in machine learning [31]. Therefore, despite the challenges associated with communication costs and training time, federated learning offers significant advantages in terms of privacy, model performance, knowledge sharing, and data ownership, making it a highly valuable and worthwhile approach to consider.

Utilizing the federated learning approach in IDSs enables data holders to participate in the learning process with no need to share data with other parties. For instance, Rahman et al. [5] presented a federated learning-based scheme for intrusion detection in IoT. In this scheme, each participant trains a neural network model on its own local training data while a central server is responsible for the aggregation of the trained models. The authors in this study compared the accuracy of the federated architecture with two other scenarios: first when all data are

shared to a centralized server, and second, when individual participant train their model without cooperation with others. Rey et al. [32] suggested using federated learning for malware detection in IoT. In this study, also the impact of malicious participants on the learning process has been examined. Preuveneers et al. [33] proposed a blockchain-based federated learning scheme in which participants have to be authenticated to join the network. This authentication process is operated on a blockchain, therefore, any participant is accountable for their actions. This mechanism enables the main server to audit the participant and trace the possible malicious actions in the learning process. Sarhan et al. [34] proposed a scheme to let multiple organizations work together in designing, training, and evaluating a robust machine learning-based network intrusion detection system. Markovic et al. [35] introduced Random Forest based on federated learning for finding security threats in big organizations. Aouedi et al. [36] suggested a semisupervised learning approach to improve the accuracy of intrusion detection systems.

Although federated learning reduces privacy concerns in the learning process, this approach might lead to some issues in terms of communication cost and learning time. In the federated learning process, in every round, first, a participant trains a local model, and then it should wait for other clients to finish their training tasks and share their local models to be aggregated by a centralized server [6]. This process continues for some rounds that can be much longer than the centralized learning mechanisms [13]. To tackle this problem one solution is to use feature selection to reduce the feature space size and consequently learning time

[37]. But, feature selection without sharing data to a centralized server is challenging because different local datasets are not necessarily statistically heterogeneous. Recently some studies have been performed on providing collaborative solutions for feature selection in federated learning. Hu et al. [38] suggested a method in which every participant utilizes an algorithm based on Particle Swarm Optimization (PSO) [39] for selecting the most useful features, and then a trusted third party is responsible for aggregating these local feature sets. Cassar et al. [40] proposed a method that uses aggregated cross-entropy for feature selection in federated learning-based Cyber-Physical Systems in autonomous vehicles. Banerjee et al. [41] suggested an algorithm that uses two scores for feature ranking in local datasets: feature-feature mutual information and feature-class mutual information. These scores are used by a central system to generate an overall ranking and generating a global dominant feature subset.

Despite the advantages of the federated learning approach in different applications, this approach increases some concerns facing adversary actions. The reason is that through federated learning, an updated version of the global model is shared among all members. This global model provides auxiliary knowledge for attackers to run poisoning and inference attacks [42]. For instance, Fang et al. [43] introduced a model poisoning attack against a Byzantine-robust federated learning architecture. The attack targets local models to compromise the global aggregated model. Bagdasaryan et al. [44] proposed a backdoor attack that generated a corrupted local model by training the model on manipulated training data. Then, this local model injects a backdoor into the global model in the aggregation phase.

The existing solutions for intrusion detection in IoT that leverage federated learning are prone to poisoning attacks. In this study, first, we depict the vulnerability of federated learning against potential attacks by designing a GAN-based poisoning attack on federated IDSs, then we propose a solution utilizing Social IoT. The presented solution is privacy-preserving and provides a trust-based mechanism for cooperation in federated learning-based IDSs.

3. GAN-based attack

Training models on federated local devices provide an opportunity for adversaries to mimic the trusted participant and perform malicious activities. Different types of attacks can be launched against federated learning architecture [10]. One type of these attacks is the model poisoning attack that recently has been utilized for targetting federated learning [44]. Model poisoning attack is an attack in which the adversary tries to mislead the global trained model in federated learning to misclassify a set of input samples with high confidence by adding a manipulated model called the poisoned model [42].

In this study, we focus on the GAN-based attack, one of the major attacks to disrupt the performance of the generated model in the federated learning process [21]. This attack has been utilized in some applications such as image processing and it has been shown that the GAN-based attacks are highly capable of compromising the generated model in the federated learning approach [23].

In this section, we design a GAN-based attack in federated intrusion detection systems to examine the learning process in presence of malicious participants that aims to degrade the performance of the general model.

Table 1 demonstrates the list of notations used in this paper.

3.1. Threat model

We consider the setting that is suggested by Nitti et al. [45] which presumes some of the local clients aim to perform malicious actions. The goal of these adversaries is to corrupt the global model by uploading poisoned models. These poisoned models compromise the global model and lead it to learn the patterns that are desired by adversaries. Using GAN-generated samples, the malicious participants have control over

Table 1

Notations.

Acronym	Description
N	The Number of Participants
P_i	The i th Participant
D_i	The i th Local Dataset
G	The Generator
D	The Discriminator
Z	Random Noise in GAN
Y	The target class
D_{Gen}	The Generated samples by the Generator
D_{Pois}	The poisoned data
M_{Pois}	The poisoned model
r_p	The rate of poisoned samples
SPR_{ij}	Social Participant Relationship similarity between P_i and P_j
OPR_{ij}	Ownership Participant Relationship similarity between P_i and P_j
SOR_{ij}	Social Owner Relationship similarity between P_i and P_j
f_i	List of the friends of P_i
C_{ij}	The centrality of P_j with respect to P_i
f^{MID}	Mutual Information Difference Function
V_i^{MID}	The vector of Mutual Information Difference values calculated by i
S	The selected features list

the generated poisoned model by adjusting the percentage of poisoned data.

The malicious participants need no training data when joining the system. Moreover, we assume that attackers work stand-alone, which means an adversary does not need to cooperate with other attackers to act.

3.2. Attacker's capability

To successfully run an attack in the federated learning process, each attacker joins as a participant in the system and has access to the general model which is shared with all participants. The adversary is an active insider that has full control over its own training procedure while is not capable of influencing the aggregation process, accessing any other benign training data, controlling the test procedure, and affecting the agreed learning algorithm. Moreover, an attacker can not steal the identity of other participants.

3.3. Attack description

In federated learning, the local models are trained on private datasets invisible to the attackers. Therefore, it is more difficult for adversaries to design an attack on the general model with no knowledge regarding the distribution of training data. A GAN model can help attackers to generate data similar to the benign participants' training data. In this study, we design a model poisoning attack that leverages GAN to generate poisoned data. These data will be used by the attackers to train poisoned models aiming to compromise the global model's performance. An overview of the attack is shown in Fig. 2.

A GAN consists of a generator and a discriminator that compete in an adversarial game [46]. The discriminator tries to distinguish the fake data produced by the generator, while the generator aims to yield samples similar to real data. In our attack scenario, the attacker has no access to the samples on the other local datasets, and thus can not use real data to train the GAN model. But the participants in the federated learning have the chance to get access to a trained global model which is trained to discriminate data samples with respect to their label. This global model can be used in the GAN architecture as the discriminator and the generator is trained based on the labels produced by this copy of the global model. The designed procedure for generating the poisoned attack is illustrated in Algorithm 1.

In this approach, the attacker utilizes a copy of the trained global model as the initialized discriminator. Then generator gets random noise as input and produces new samples that have the same size as the real

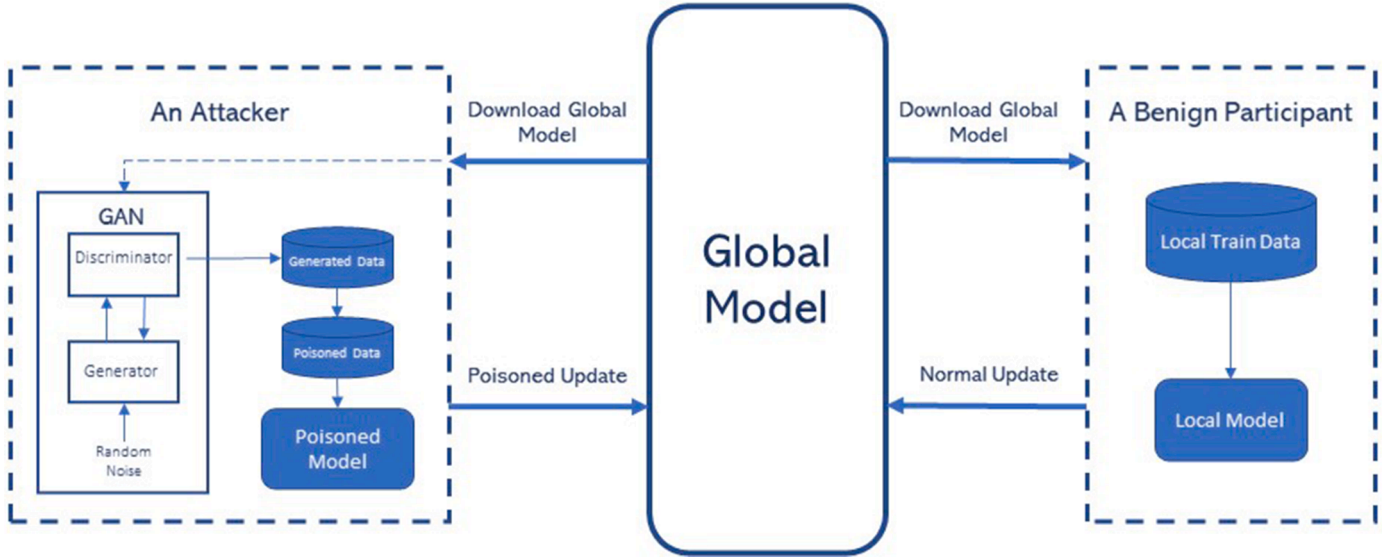


Fig. 2. An overview of GAN-based poisoning attack in federated learning architecture.

Algorithm 1

GAN-based poisoning attack.

Input: Global Model (M_{Global})
 Target Label (Y)
 Output: A poisoned Model

- 1: $G \leftarrow$ Initialize generator
- 2: $D \leftarrow M_{Global}$
- 3: for $i = 1, 2, \dots$
- 4: $Z \leftarrow$ Generate noise samples by G ...
- 5: for sample $x_i \in Z$
- 6: if $D(x_i)$ is not Y
- 7: update G
- 8: end if
- 9: end for
- 10: end for
- 11: $D_{Gen} \leftarrow$ New samples generated by G
- 12: $D_{Pois} \leftarrow$ poisoning r_p percent of D_{Gen} by changing the labels
- 13: M_{pois} is trained using D_{Pois}

end procedure

data, then sends these samples to the discriminator. If the discriminator classifies a sample as benign, it will be presumed as *real*, otherwise, the sample receives the *fake* label, and also the generator receives feedback to be updated. The process continues until the generator reaches a desired level of accuracy. After this step, the generator is trained to produce samples that can be detected as benign by the general trained model.

In the next step, the attacker uses the generator to yield benign data samples and deliberately disturbs data by label changing. Label changing means assigning an attack label to the sample. Having such poisoned data enables the attacker to generate a poisoning model to compromise the global model in the next federated learning rounds. By adjusting the rate of changed labels the attacker is capable of launching controlled attacks which make it difficult for the main server to detect the poisoned models.

3.4. Poisoning attack

In this section, we execute some experiments to examine the effectiveness of the introduced attack in compromising the global model. In the poisoning attack evaluation, we consider the effect of sharing poisoned models by malicious participants. To this end, we designed a GAN in which the utilized generator is a 3-layer Dense neural network. The first layer has 32 neurons, the second layer consists of 64 neurons,

and the number of neurons in the last layer is equal to the number of features. The Discriminator is a Convolutional Neural Network (CNN) consisting of two convolution layers (256 and 128 neurons) and two dense layers (128 and 1 neurons) and each convolution layer is followed by a max-pooling and a dropout layer.

In our scenario, first, the adversaries wait for ten rounds until the global model is trained. Then a malicious participant utilizes this global model in a GAN architecture to generate new samples and poison some of these samples by changing the labels. In the next step, a poisoned model is trained over these data and is aggregated with other models. Fig. 3 illustrates the effect of poisoning models on the global model's performance considering different numbers of adversaries added to the ten benign participants. Fig. 3(a) presents the Loss value in the presence of adversaries while Fig. 3(b) and (c) consider Accuracy and F1-Score respectively. As it can be seen in the figures, in the first ten rounds when no poisoned models are contributing to the aggregation, the loss is decreasing while accuracy and F1-Score are increasing.

Adding an adversary can slightly decrease the performance of the model while increasing the number of adversaries intensifies the destructive effect. It is worth mentioning that in the designed experiments the adversaries work independently, and they do not need to work with or even be aware of other malicious participants in the system. This shows that federated learning is highly vulnerable when it is launched in uncertain environments in which untrusted participants can join and collaborate in the learning process. The major reason that allows adversaries to run poisoning attacks against federated learning is that all participants are presumed to be trustable. Such a presumption is not rational in real-world applications. In Section 4, we propose a trust-oriented privacy-preserving intrusion detection system for IoT that prevents untrusted participants from participating in the learning process.

4. Proposed approach

4.1. Problem definition

We assume that there are N participants in the system that work together to train a machine learning model in a federated manner. A participant can be an individual IoT device with enough computation capacity to train a local model (e.g., smartphone) or an edge server that represents a group of IoT devices. Our goal is to build a global model that leverages the collaboration of the participants to provide higher performance than the individual local models. Here we aim to protect the

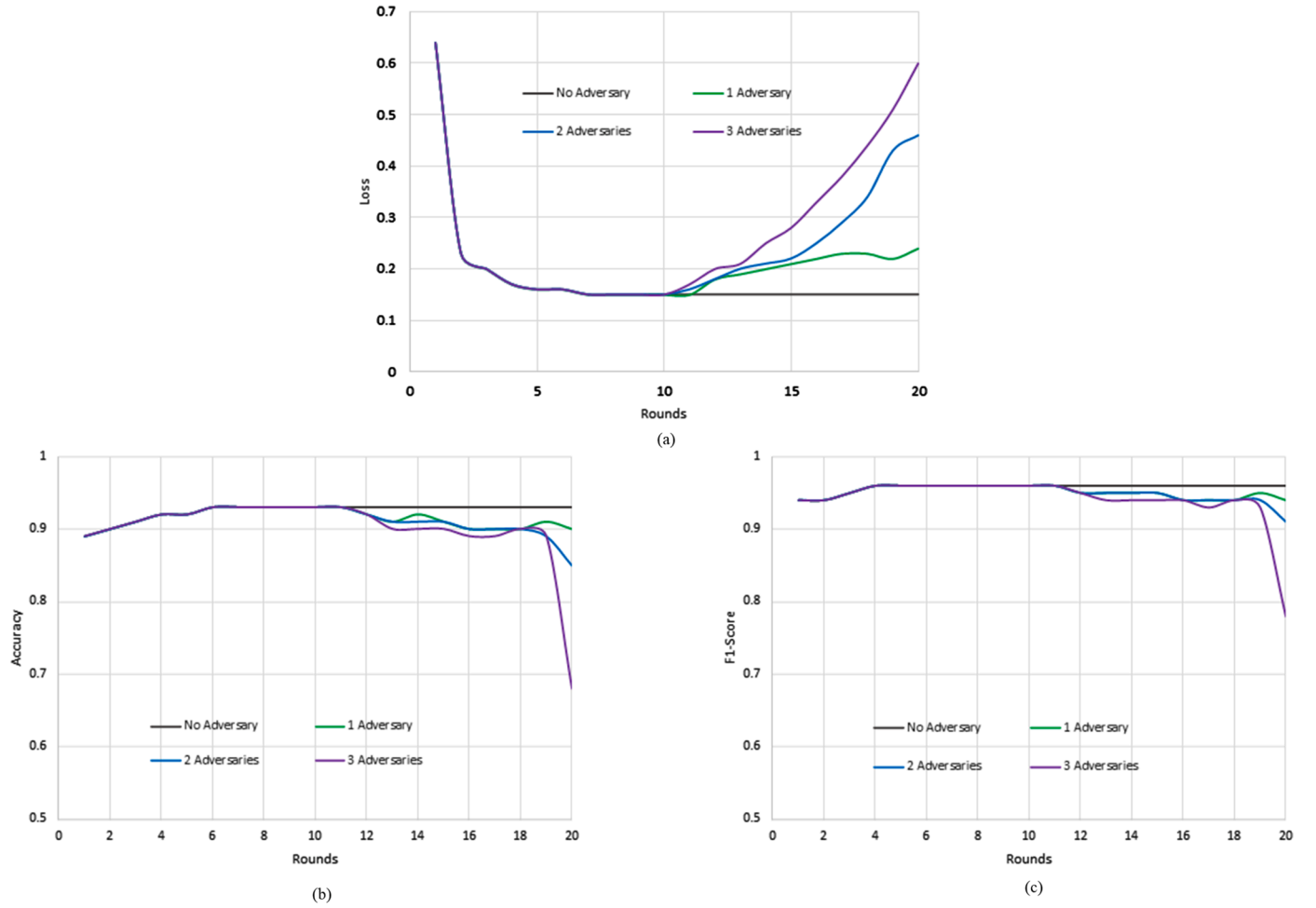


Fig. 3. Performance of the global model against the poisoning attack. (a) Loss (b) Accuracy (c) F1-Score.

privacy of the local datasets and provide a trust-based environment for collaborative learning.

The participants are denoted by $\{P_1, P_2, \dots, P_N\}$ in this article and each one has a local training dataset represented by $\{D_1, D_2, \dots, D_N\}$. In each round, the participant P_i trains a local model M_i on its local dataset D_i and after each round, these local models are aggregated to build a global model. We assume the local datasets have the same feature set while they may be different in terms of volume and distribution.

4.2. Trust-based federated learning for IoT

In federated learning-based solutions, a major challenge is to detect the model attacks that aim to compromise the global generated models [42]. The distributed and heterogeneous nature of data in federated learning makes it challenging to find these attacks. With the learning process distributed across multiple devices, monitoring and analyzing the training process becomes inaccessible. Moreover, the data used by different participants can vary significantly in terms of size, distribution, and quality which makes it difficult to detect malicious activities. The diversity of data introduces complexities in distinguishing between genuine variations and potential compromising attacks [47]. Consequently, specialized techniques are needed to ensure the security and integrity of the federated learning process.

Recent studies have shown the capability of the Social IoT to address challenges in uncertain environments [18]. In Social IoT, the IoT devices are assumed to be able to establish social relationships based on the social connection of their owners [48]. Leveraging this concept, IoT devices can build a reliable friendship network in which only trusted

entities are included [19].

In this study, we propose a trust-based federated learning approach that leverages the Social IoT notion to provide a robust privacy-preserving collaborative learning mechanism for intrusion detection in uncertain IoT environments. This approach helps the participants to eliminate the untrusted participants from the collaborative learning process and increases the robustness of the system against adversaries. Moreover, participants can adjust the impact of each of their friends in the learning process according to the closeness of their social relationships. In addition, the proposed approach does not need a central server to handle the collaboration and perform the aggregation, because the participants manage the learning process by themselves.

According to Social IoT specifications [19], each IoT device has an owner that might own more than one entity in the system. We assume that participants can build three types of social relationships with others:

- *Social Participant Relationship (SPR)*: Every participant has a list of other participants that are socially connected to them. These connections might be defined based on the location, application, or missions followed by the participants. The trustability of these participants is evaluated by a trust management system [49].
- *Ownership Participant Relationship (OPR)*: The participants that belong to the same owner make a community in which the members build this type of social relationship.
- *Social Owner Relationship (SOR)*: A social connection that is defined among owners. According to this relationship, each owner also has a list of other owners as friends.

Based on these types of social relations, we use the notion of Social Tie (ST) [48] as a metric to evaluate the level of social connections between two participants in SIoT as follows:

$$ST_{ij} = (\alpha \times SPR_{ij}) + (\beta \times OPR_{ij}) + (\gamma \times SOR_{ij}) \quad (1)$$

In this formula, ST_{ij} is the Social Tie evaluation of P_i regarding P_j . SPR_{ij} is Jaccard Similarity [50] of Participant Friendship lists of participant i (SPR_i) and participant j (SPR_j); while OPR_{ij} and SOR_{ij} are the Jaccard Similarity for Ownership Participant Relationship and Social Owner Relationship of the two participants, respectively. Parameters α , β , γ are the weight of each factor in the formula. These values are real numbers between [0,1] and their summation is equal to one.

Using the Social Tie metric, each participant is able to build a list of close friends in which the participants with an ST value higher than a threshold are included. This threshold might vary for each participant and is set by IoT device owners. Fig. 4 demonstrates the process of selecting close friends for federated learning. In Fig. 4(a), five different owners are depicted. Using Eq. (1) participant i adds 9 other participants to its close friend list, and participant j finds 11 close friends while 7 participants are mutual in both lists.

In the learning procedure, each local server can leverage the collaboration of its trusted close friends. This helps to improve the performance of the learning model and limits the untrusted participants to engage in the learning process. Fig. 4(b) shows this collaboration. A local server (e.g., server m) starts the process. We name this node as master. First, the master sends a message to its friend asking which ones have samples of the desired data and are willing to participate in the learning process. Then the master initializes a training model (M^0) to be shared among the other volunteer participants. In the next step, the participants download the model and train it on their local training datasets (D_i) and return the local model (M_i) back to the master. After that, the master node aggregates the local models once all the local models are received leveraging the notion of centrality [45]. The centrality of P_j with respect to P_i represents how much P_j is central in the friendship network of P_i and is defined as follows:

$$C_{ij} = \frac{|f_i \cap f_j|}{|f_i|} \quad (2)$$

This formula calculates C_{ij} as the number of mutual friends of P_i and P_j by the number of P_i 's friends. After collecting all local models, the master aggregates these models using weighted averaging while every local model is weighted based on its centrality to the master node. The proposed aggregation formula is:

$$w_{Global}^k = \frac{\sum_{i=1}^N C_{mi} w_i^r}{\sum C_{mi}} \quad (3)$$

In which w_i^r is the vector of local parameters of M_i in the round r and is calculated as follows:

$$w_i^r = w_i^{r-1} - \eta L(w_i^{r-1}) \quad (4)$$

In this study, we utilize the cross-entropy loss function [30] as:

$$L(w) = \sum_{i=1}^C p(y=i) E_{x|y=i} [\log f_i(x, w)] \quad (5)$$

In which C denotes the classes, y is the predicted label, and f_i is the probability for the i th class parametrized over w .

The learning process continues for some rounds and finally, the master can replace its own local model with the global model. The learning process of the proposed approach is described in Algorithm 2.

4.3. Collaborative feature selection for SIoT

Feature selection fastens the learning process by eliminating data features that are irrelevant or less relevant to the model's goal. In federated learning architecture, the presence of heterogeneous and

Algorithm 2

The learning procedure of the proposed approach.

Comment: m denotes the master node

Input: The list of friends of m ,

The list of centrality values of friends of m

Output: The global model

1: $P \leftarrow$ The list of willing friends

2: $N \leftarrow$ The number of willing friends

3: Initialize M^0

4: for each round $r = 1, 2, \dots$:

5: for each i in P (in parallel):

6: $w_i^{r-1} = w_{Global}^{r-1}$

7: $B \leftarrow$ Splitting D_i into batches size B

8: for each epoch $e = 1, 2, \dots$:

9: for each batch b in B :

10: $w_i^r \leftarrow w_i^{r-1} - \eta L(w_i^{r-1})$

11: end for

12: end for

13: end for

14: $w_{Global}^k \leftarrow \frac{\sum_{i=1}^N C_{mi} w_i^r}{\sum C_{mi}}$

15: end for

end procedure

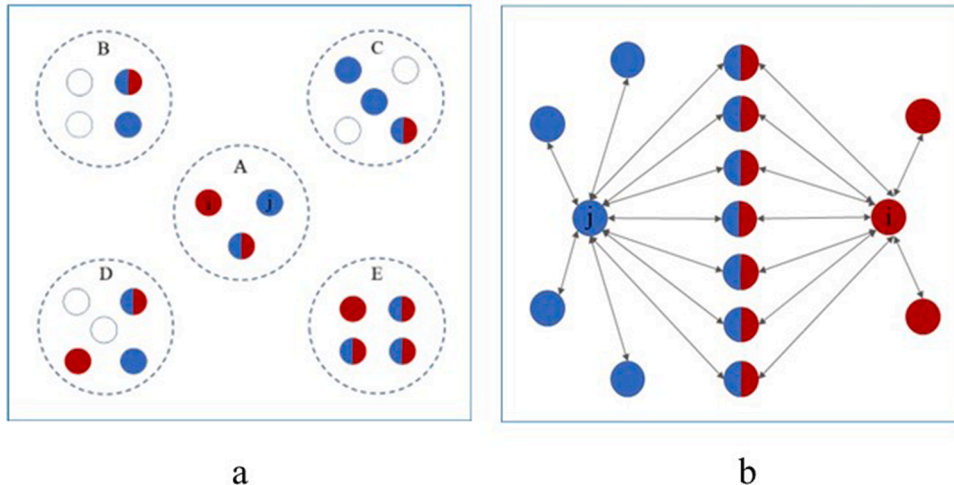


Fig. 4. The process of selecting trusted close friends and social collaboration.

potentially non-overlapping features across clients creates a complex challenge to choose the best features for the learning process. Addressing this problem requires the development of novel methods that can handle disparate features, adapt to varying data distributions, and effectively aggregate information. An effective solution for these challenges should be capable of identifying discrepancies within feature spaces and consolidating diverse feature importance from various participants' viewpoints. In this section, we propose a collaborative method that enables the IoT participants to leverage social collaboration from their trusted close friends to build a mutual list of selected features. The obtained results show the effectiveness of the presented method (See Section 5). This method also enhances the performance of feature selection while preserving the privacy of the training data because the participants do not need to reveal any data samples to others.

The proposed collaborative method leverages the Minimum Redundancy Maximum Relevance (MRMR) [51], a filter feature selection method that recently has gained attention and has been used in different studies. This method aims to select the relevant features while controlling redundancy within the chosen features.

The objective function of the MRMR is to iteratively add features with the maximum value of Mutual Information Difference (MID) that is calculated as follows [51]:

$$f^{MID}(X_i) = I(Y, X_i) - \frac{1}{|S|} \sum_{X_s \in S} I(X_s, X_i) \quad (6)$$

In this formula, S is the set of selected features, X_s is a selected feature, X_i denotes a feature that has not been selected yet, and Y is a target label. I is the mutual information function and is defined as follows [52]:

$$I(Y, X_i) = \sum_{x \in X_i} \sum_{y \in Y} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (7)$$

In which p is the marginal density function [52].

The proposed social collaborative feature selection process is outlined in Algorithm 3. To begin with, the master node initializes an empty list of selected features and proceeds to share them with its trusted close friends who have expressed their willingness to participate in the process. Each participant gets the list, calculates V^{MID} , a vector of calculated $f^{MID}(X_i)$ for all features on the local dataset, and subsequently transmits the vector back to the master. The master takes this information as a recommendation from other participants and takes this into account in the calculations considering the level of trustability of the participants. To this end, the master first compares every received vector to its own feature space and keeps the values for the mutual features then calculates a weighted average of the vectors. The weights of each vector are determined based on the Centrality value calculated in Eq. (2). In the

Algorithm 3

Collaborative feature selection.

Comment: m denotes the master node
Input: The list of friends of m ,
The list of centrality values of friends of m
The number of features to be selected(K)
Output: A list of the selected features
1: $P \leftarrow$ The list of willing friends
2: $N \leftarrow$ The number of willing friends
3: $S \leftarrow \emptyset$
4: for $k = 1, 2, \dots, K$
5: for each i in P (in parallel):
6: $V_i^{MID} \leftarrow$ The vector of calculated MID values
7: end for
8: $V_{Avg}^{MID} \leftarrow \frac{\sum_{i=1}^N C_{mi} V_i^{MID}}{\sum_{i=1}^N C_{mi}}$
9: $s \leftarrow \text{argmax}(V_{Avg}^{MID})$
10: $S.add(s)$
11: end for
end procedure

next step, the master participant selects a feature with the maximum weighted averaged values of f^{MID} and adds it to the list of the selected features. This procedure iterates until the desired number of features is chosen. The proposed feature selection mechanism leverages the advantages of the MRMR algorithm in selecting features with maximum relevance and minimum redundancy. Moreover, it provides a privacy-preserving mechanism for social collaborative feature selection within a federated architecture.

5. Evaluation

5.1. Dataset

To evaluate the proposed approach, we used the UNSW-NB15 dataset [53]. This dataset simulates a modern network environment by launching some contemporary types of attacks. We utilized this dataset in the binary format which means every sample has a *Benign* or *Attack* label. The original dataset contains around 2 and a half million records with 48 features in total. These features are listed in Table 2.

To utilize the dataset in a federated scenario, we presumed every individual IP represents a different participant; therefore, the samples are distributed based on the destination IP address of the network packets. We selected the participants that have a least one attack sample in their local dataset. In the UNSW-NB15 dataset, ten different nodes have these characteristics. An overview of the samples which belong to these ten participants is demonstrated in Table 3.

For social relationship data, we adopted real-world data from a social media outlet to simulate social connections for SIoT. The utilized data are extracted from the ego-Twitter dataset provided by the Stanford Network Analysis Project [54]. We picked one ego network containing ten nodes therefore the number of participants is the same as the number of participants in the experiment. Table 4 describes the social connection data that are needed for Social Tie calculation. It also has been assumed that the first and the third owners are friends.

5.2. Evaluation metrics

In machine learning, the performance of a model is described based on the comparison of the predicted results of a trained model with the actual values. To evaluate the effectiveness of the proposed approach, we brought some metrics to the account; namely, Loss, Accuracy, and

Table 2

Features list of UNSW-NB15.

No.	Feature Name	Feature Type	No.	Feature Name	Feature Type
f_1	srcip	nominal	f_{25}	trans_depth	numeric
f_2	sport	numeric	f_{26}	res_bdy_len	numeric
f_3	dstip	nominal	f_{27}	Sjit	numeric
f_4	dsport	numeric	f_{28}	Djit	numeric
f_5	proto	nominal	f_{29}	Stime	timestamp
f_6	state	nominal	f_{30}	Ltime	timestamp
f_7	dur	numeric	f_{31}	Sintpkt	numeric
f_8	sbytes	numeric	f_{32}	Dintpkt	numeric
f_9	dbytes	numeric	f_{33}	tcprtt	numeric
f_{10}	sttl	numeric	f_{34}	synack	numeric
f_{11}	dttl	numeric	f_{35}	ackdat	numeric
f_{12}	sloss	numeric	f_{36}	is_sm_ips_ports	binary
f_{13}	dloss	numeric	f_{37}	ct_state_ttl	numeric
f_{14}	service	nominal	f_{38}	ct_flw_http_mthd	numeric
f_{15}	Sload	numeric	f_{39}	is_ftp_login	binary
f_{16}	Dload	numeric	f_{40}	ct_ftp_cmd	numeric
f_{17}	Spkts	numeric	f_{41}	ct_srv_src	numeric
f_{18}	Dpkts	numeric	f_{42}	ct_srv_dst	numeric
f_{19}	swin	numeric	f_{43}	ct_dst_ltm	numeric
f_{20}	dwin	numeric	f_{44}	ct_src_ltm	numeric
f_{21}	stcpb	numeric	f_{45}	ct_src_dport_ltm	numeric
f_{22}	dtpcb	numeric	f_{46}	ct_dst_sport_ltm	numeric
f_{23}	smeansz	numeric	f_{48}	ct_dst_src_ltm	numeric
f_{24}	dmeansz	numeric			

Table 3

Distributed local training data.

Local Dataset	Benign Samples	Attack Samples	Total
1	38,082	2911	40,993
2	8231	5008	13,239
3	25,812	2497	28,309
4	11,719	4826	16,545
5	39,007	3429	42,436
6	43,754	3741	47,495
7	6931	4609	11,540
8	21,080	3991	25,071
9	92,991	2671	95,662
10	11,195	3839	15,034

Table 4

Social connection data.

Participant	Friend Participants	Owner
1	9, 10	1
2	1, 5, 8	1
3	4, 5	1
4	3, 5, 9, 10	1
5	1, 3, 4, 8	2
6	7, 8	2
7	1, 8	2
8	9, 10	3
9	8, 10	3
10	1, 5, 9	3

F1-Score. Loss is the mean error of predicted results across all testing samples. Accuracy and F1-Score are calculated using True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). TF indicates intrusions that are successfully detected by the model. FP shows the number of benign samples incorrectly classified as intrusion. TN represents the number of benign records truly labeled. FN indicates the number of intrusions that are classified as benign. Utilizing these metrics, Accuracy, and F1-Score are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$F1 - Score = \frac{2 \times TP}{2 \times TP + (FP + FN)} \quad (7)$$

5.3. Experimental setup

We implemented the proposed method using the TensorFlow framework. In each round of training, each local model is trained independently then these models will be shared with a master node to be aggregated. We utilized the Convolutional Neural Network (CNN) architecture to construct the classifier. This model consists of an input layer, two convolution layers, and two dense layers while each convolution layer is followed by a max-pooling and a dropout layer. The number of neurons in each layer is determined using Grid Search [55] which are 48, 256, 128, 128, and 1, respectively. Using Grid Search, we also selected the *relu* activation function and set the learning rate as 0.001. The initial weights of the models are set randomly, and the threshold for Social Tie is set as 0.5. Moreover, as mentioned in Section 5.1, we have 10 independent IoT participants.

We evaluated the performance of the trained model in two different scenarios. We call the first scenario self-learning in which each IoT node trains a model independently using its own local data. The second scenario utilizes SIDS, the proposed model, in which the IoT nodes leverage collaboration from their trusted local friends.

5.4. Experimental results

5.4.1. Other federated learning-based solutions performance

Recently, some approaches have been proposed for utilizing federated learning for intrusion detection. Table 5 presents a comparison among multiple studies that were implemented in a similar environment and tested using the same dataset as the current study. All of these studies were designed in distributed environments, where the learning process was conducted using the federated learning approach. Furthermore, they were all tested on the UNSW-NB15 dataset. The difference among these studies lies in the machine learning models they have used.

Sarhan et al. [34] proposed a federated deep neural network for intrusion detection in IoT. The presented solution achieved 0.91 on both accuracy and F-Score metrics. Markovic et al. [35] suggested a random forest-based federated learning solution for intrusion detection systems. The presented method in this study obtained an accuracy of 0.79. Aouedi et al. [36] introduced a federated semi-supervised approach to detect intrusions in a system. To evaluate their approach, the author of this paper focused on the F-Score metric. They reported values of 0.86 for benign instances and 0.8 for attack instances. In another study, Bertoli et al. [56] suggested using federated autoencoders for intrusion detection in IoT. The presented solution attained an accuracy of 0.97, along with F-Score values of 0.73 for benign instances and 0.66 for attack instances. It is worth mentioning that none of these studies considered the possibility of model attacks. However, in Section 3, we demonstrated the vulnerability of these approaches to compromising attacks. In this study, we evaluated the performance of the proposed method in the presence of adversaries that aim to execute compromising attacks. In order to facilitate a fair comparison with prior research, we designed the experiments following the framework suggested by Sarhan et al. [34] in their study.

5.4.2. SIDS model performance

This section presents the evaluation results of the proposed approach. Table 6 shows the performance of the proposed model (SIDS) in comparison to the self-learning approach and federated learning approach. In the self-learning scenario, each participant works independently and does not use assistance from others. In the federated learning scenario, we adopted the experimental design proposed by Sarhan et al. [34]. In addition to the regular participants (10 participants), we introduced an assumption where three adversaries attempt to compromise the general model by sharing poisoned models. Similar to other participants, the friendship list of the malicious participants is obtained from the ego-Twitter dataset [54] in which the friends for the first adversary are nodes 2, 3, 6, 9; for the second adversary the friends are the nodes 3, 5; and the friends of the last adversaries are 1, 5, 7. Each malicious participant is presumed to have a distinct owner.

In the last scenario, the proposed model is used for model training. Since the presented method selects the participants for the collaborative learning process, the number of participants in the learning process is fewer than when all participants in the system are presumed trustable. Using fewer participants might lead to losing some recommendations from the other participants. However, it is not reasonable to rely on such a presumption and consider all participants trustable in real-world applications. The notion of SIoT helps us to provide a community for each participant in which the participants can collaborate based on the level

Table 5

Compare other federated-learning-based solutions.

Paper	Accuracy	F-Score
Sarhan et al. [34]	0.91	0.91
Tarkovic et al. [35]	0.79	–
Aouedi et al. [36]	–	0.86/0.8 (benign/attack)
Bertoli et al. [56]	0.97	0.73/0.66 (benign/attack)

Table 6

Evaluation results of self-learning and federated learning approaches in SIoT using all data features.

Participant	Self-Learning			Federated Learning			SIDS		
	Loss	Accuracy	F1-Score	Loss	Accuracy	F1-Score	Loss	Accuracy	F1-Score
1	0.25	0.88	0.93	0.41	0.76	0.79	0.24	0.89	0.94
2	0.72	0.43	0.57	0.67	0.43	0.56	0.32	0.8	0.88
3	0.22	0.91	0.94	0.51	0.72	0.73	0.31	0.87	0.9
4	0.55	0.68	0.78	0.44	0.71	0.75	0.31	0.86	0.92
5	0.37	0.82	0.89	0.51	0.73	0.77	0.29	0.87	0.92
6	0.34	0.8	0.88	0.45	0.74	0.76	0.41	0.83	0.89
7	0.75	0.51	0.65	0.62	0.51	0.64	0.26	0.87	0.92
8	0.23	0.91	0.95	0.48	0.73	0.78	0.42	0.8	0.87
9	0.77	0.71	0.8	0.48	0.73	0.77	0.39	0.77	0.84
10	0.33	0.89	0.93	0.53	0.7	0.74	0.37	0.85	0.89
11	–	–	–	0.89	0.51	0.54	0.89	0.51	0.54
12	–	–	–	0.95	0.47	0.49	0.95	0.47	0.49
13	–	–	–	0.73	0.61	0.79	0.73	0.61	0.79
Average (Attackers Excluded)	0.45	0.75	0.83	0.50	0.67	0.72	0.33	0.84	0.90

of friendship and trustworthiness.

As is shown in Table 6, in seven of the ten participants the proposed solution enhances both Accuracy and F1-Score comparing two other scenarios. The most improvement belongs to participants 2 and 7. On the other hand, among the participants whose performance is reduced, in participants 3 and 10, the decrement in both metrics is less than 5 %. Comparing the average values for performance metrics in the three scenarios shows an overall improvement in all metrics. The proposed method improves the average accuracy by 9 % and 17 % comparing the self-learning and federated scenarios respectively; and enhances the average F1-Score by 7 % and 18 %; while reducing the loss value by 0.12 and 17.

5.4.3. Feature selection

The goal of feature selection is to accelerate the learning process by removing the features that are less relevant to class labels and minimizing the correlation among the selected features. To assess the effect of feature selection we used two scenarios. In the first scenario, we assumed the local nodes work standalone without using the collaboration of other IoT devices. Thus we applied the MRMR algorithm individually for each of the local datasets to extract a reduced list of features. In the filtering process, 20 of the best features out of the 48 features presented in Table 2 are selected for every individual participant. Table 7 demonstrates the lists of the selected features. The table shows the selected feature sets by different participants differ significantly because each participant has no information on the most useful features from other participants' perspectives and merely relies on its own local available data. Fig. 5 shows the frequency of the used features by the participants. As can be in the figure, only three features are selected by all participants, while 37 features (77 % of all features) are used by at least one participant. The result shows a major challenge for this approach that is to reach a final feature set most useful for the entire learning process.

Table 7

The lists of features chosen through independent feature selection.

Participant	Selected Features
1	$f_6, f_{11}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{29}, f_{30}, f_{33}, f_{34}, f_{35}, f_{37}, f_{38}, f_{40}, f_{42}, f_{44}, f_{45}, f_{46}, f_{47}$
2	$f_7, f_{10}, f_{14}, f_{16}, f_{17}, f_{19}, f_{23}, f_{24}, f_{29}, f_{30}, f_{33}, f_{34}, f_{35}, f_{36}, f_{39}, f_{40}, f_{41}, f_{42}, f_{43}, f_{47}$
3	$f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{32}, f_{33}, f_{34}, f_{35}, f_{37}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47},$
4	$f_5, f_6, f_{14}, f_{16}, f_{19}, f_{20}, f_{21}, f_{22}, f_{24}, f_{25}, f_{27}, f_{33}, f_{34}, f_{35}, f_{38}, f_{39}, f_{41}, f_{42}, f_{44}, f_{46}$
5	$f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{30}, f_{33}, f_{34}, f_{35}, f_{37}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
6	$f_5, f_6, f_{10}, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
7	$f_{10}, f_{11}, f_{14}, f_{16}, f_{17}, f_{23}, f_{24}, f_{25}, f_{28}, f_{33}, f_{34}, f_{37}, f_{39}, f_{40}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{47}$
8	$f_5, f_7, f_{10}, f_{11}, f_{19}, f_{20}, f_{21}, f_{22}, f_{29}, f_{30}, f_{31}, f_{33}, f_{34}, f_{35}, f_{40}, f_{41}, f_{42}, f_{43}, f_{44}, f_{46}$
9	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{27}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
10	$f_5, f_6, f_{10}, f_{11}, f_{15}, f_{16}, f_{19}, f_{20}, f_{21}, f_{22}, f_{25}, f_{27}, f_{32}, f_{33}, f_{34}, f_{35}, f_{38}, f_{41}, f_{42}, f_{46}$

Next, we utilized the collaborative feature selection described in Section 4.3. Utilizing the social relationship characteristics of Social IoT, in collaborative feature selection an individual participant generates a list of the best feature by leveraging recommendations from other participants considering the level of friendship closeness. The lists of the selected features through this approach are shown in Table 8. As can be seen in the table, many of the selected features are the same for different participants. The frequency of the selected features by different participants is demonstrated in Fig. 6. The figure shows out of twenty features used by each participant, seventeen are the same. The reason is the proposed method enables participants to gain an overview of the most useful features of all local training data through social collaboration, even though this method does not require direct access to other local datasets. Using this method, the participants gain a common overall understanding of the most useful features resulting in similar selected feature sets.

In the next step, we examined the impact of the presented collaborative feature selection method on the learning process. To this end, we selected 20 of the most useful features and ran some experiments using three scenarios. The first scenario is self-learning in which the participants work independently in feature selection and learning phases. In the second scenario, the master node independently decides about the selected features and then runs the proposed approach in Section 4.2 for collaborative learning. In the third scenario, the proposed collaborative feature selection is utilized alongside the SIDS. As can be seen in Table 9, the proposed collaborative feature selection method outperforms the other scenarios. The most improvement in the second approach belongs to participant 2 while in the third scenario, participants 4 and 7 have the most performance enhancement. Considering the overall values performance metrics, in the third scenario, the accuracy is 6 % more than in the first scenario and 2 % higher than in the second scenario, also, F1-Score is improved by 6 % and 2 % compared to the first and second scenarios respectively.

6. Discussion

In this study, we have demonstrated the effectiveness of our proposed model in enhancing intrusion detection mechanisms through collaborative learning. SIDS is shown to maintain good performance in the presence of adversaries launching compromising attacks. The success of SIDS over regular federated learning approaches when facing attacks is attributed to the introduced Social Tie (ST) measurement, which fosters trust-based collaboration among participants. This trust-based environment empowers IoT stakeholders to participate in IDSs without compromising data privacy. Moreover, SIDS outperforms self-learned models due to its unique advantage of gaining a general understanding of the distribution of training data, even without direct

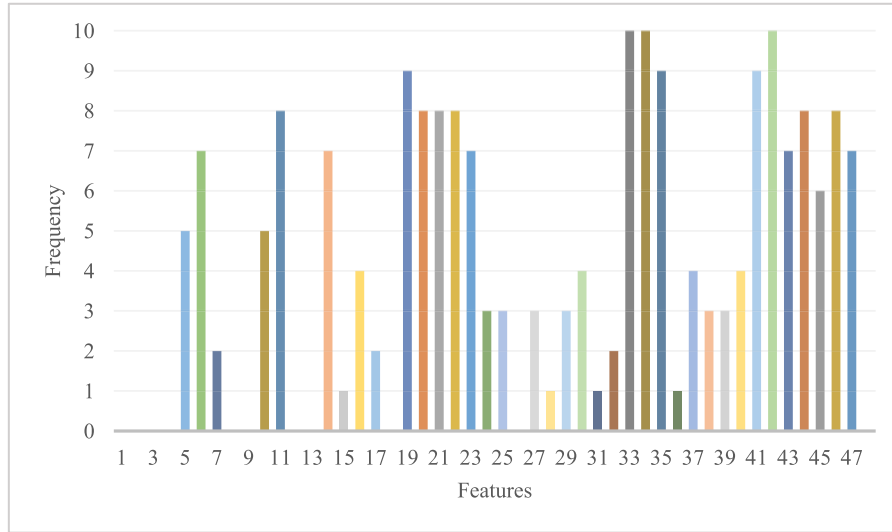


Fig. 5. The frequency of used features by the participants through independent feature selection.

Table 8

The lists of features chosen through collaborative feature selection.

Participant	Selected Features
1	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{31}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
2	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{29}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
3	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{27}, f_{30}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{46}, f_{47}$
4	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{29}, f_{31}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{46}, f_{47}$
5	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{27}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
6	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{33}, f_{34}, f_{35}, f_{37}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
7	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{25}, f_{29}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{44}, f_{45}, f_{46}, f_{47}$
8	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{27}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
9	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{30}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$
10	$f_5, f_6, f_{11}, f_{14}, f_{19}, f_{20}, f_{21}, f_{22}, f_{23}, f_{31}, f_{33}, f_{34}, f_{35}, f_{41}, f_{42}, f_{43}, f_{44}, f_{45}, f_{46}, f_{47}$

access to all datasets. This is achieved through leveraging locally trained models and sharing information cooperatively. The main strength of our proposed model is evident in scenarios where some participants have limited access to training data, which may lack certain intrusion types. In such cases, participants can benefit from other local models' knowledge through the cooperative learning process, enabling them to effectively detect previously unseen malicious actions.

In addition, this paper proposes a feature selection method that leverages Social IoT. As is demonstrated in the findings, the presented method outperforms the other solutions. It is because this method provides a mutual understanding of the features that best describe the patterns in training data. Meanwhile, comparing the results from the global model trained over full feature training data with the global model that used feature-selected training data, performance reduction in both accuracy and F1-Score performance metrics can be seen. We recommend using the feature selection in situations where participants have limited computation resources available, or when reducing the training time is a priority.

Currently, a major issue in Social IoT research is the limited available data. Although SIOT can address a wide range of problems in IoT, still the publicly available data sources that can be utilized by researchers are limited. More data sources specifically generated for SIOT, can facilitate studies in this area and allow researchers to leverage ideas from SIOT to improve solutions in IoT.

7. Conclusion

In this study, we proposed a privacy-preserving approach for

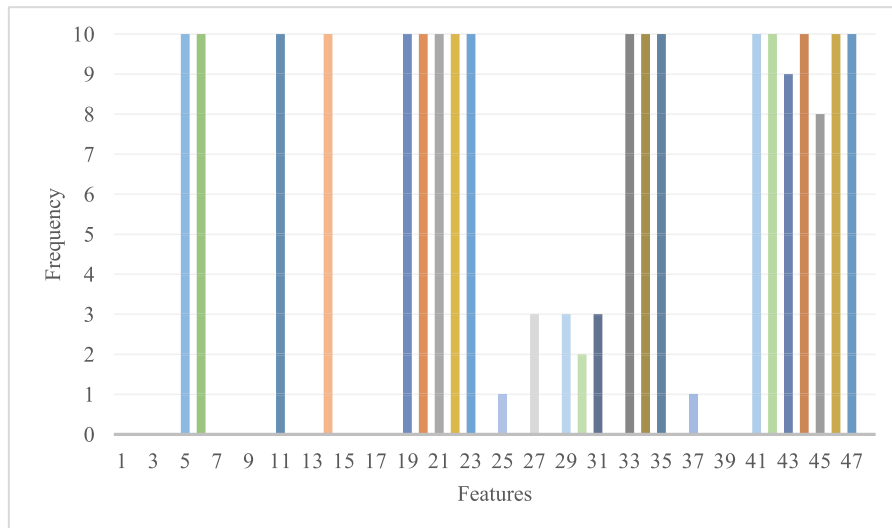


Fig. 6. The frequency of used features by the participants through collaborative feature selection.

Table 9

Evaluation results of independent and collaborative approaches in SIoT.

Participant	Self-Learning (Independent Feature Selection)			SIDS (Independent Feature Selection)			SIDS (Collaborative Feature Selection)		
	Loss	Accuracy	F1-Score	Loss	Accuracy	F1-Score	Loss	Accuracy	F1-Score
1	0.28	0.88	0.93	0.38	0.77	0.85	0.35	0.79	0.86
2	0.78	0.39	0.54	0.35	0.78	0.86	0.43	0.75	0.84
3	0.32	0.84	0.9	0.41	0.76	0.84	0.34	0.79	0.86
4	0.78	0.5	0.63	0.35	0.79	0.86	0.26	0.85	0.91
5	0.41	0.81	0.88	0.39	0.74	0.83	0.26	0.86	0.91
6	0.54	0.88	0.93	0.35	0.79	0.86	0.36	0.78	0.85
7	0.91	0.42	0.57	0.35	0.78	0.86	0.31	0.82	0.88
8	0.32	0.86	0.92	0.36	0.78	0.85	0.39	0.76	0.84
9	0.95	0.86	0.92	0.37	0.76	0.84	0.38	0.77	0.85
10	0.42	0.83	0.9	0.39	0.75	0.84	0.35	0.79	0.86
Average	0.57	0.73	0.81	0.37	0.77	0.85	0.34	0.79	0.87

detecting intrusions in SIoT. Although machine learning solutions have been widely used for IDS, these solutions have faced some serious challenges. On one hand, gathering all training data in a server to perform centralized learning increases the concerns regarding data privacy. On the other hand, collaborative learning provides a fortune for dishonest participants to compromise the aggregated model through malicious actions. In this paper, first, we designed a poison model attack on the federated learning architecture to show the vulnerability of the architecture. Then we proposed an approach that utilizes social relationship measures and provides a privacy-preserving collaborative mechanism for intrusion detection in IoT. We also suggested a collaborative feature selection method for accelerating the learning process. Our evaluation results demonstrate that the proposed model can increase both accuracy and F-Score comparing the self-learning and federated approaches while reducing the chance of untrusted participants disrupting the learning process. The evaluation results also demonstrate that the proposed collaborative feature selection mechanism outperforms the other feature selection approach in which features are selected by independent IoT devices. For future work, we aim to utilize Explainable AI techniques to improve SIDS. These techniques provide an explanation of why a behavior in IoT is detected as an intrusion. This explanation sheds light on the malicious action patterns and helps to improve security protection practices.

CRedit authorship contribution statement

Mohammad Amiri-Zarandi: Funding acquisition, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing. **Rozita A. Dara:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Xiaodong Lin:** Conceptualization, Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This research was funded by a Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant awarded to R.

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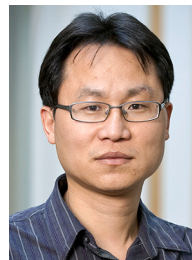
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